AITS
ALL INDIA TEST SERIES

Dropper JEE Main

AJPM8/07

Test- 07

DURATION : 180 Minutes

DATE : 09/02/2025

M.MARKS : 300

ANSWER KEY

PHYSICS

1. (4)
2. (3)
3. (1)
4. (2)
5. (2)
6. (4)
7. (4)
8. (1)
9. (4)
10. (2)
11. (3)
12. (1)
13. (2)
14. (4)
15. (2)
16. (3)
17. (1)
18. (1)
19. (1)
20. (2)
21. (4)
22. (36)
23. (37)
24. (27)
25. (7)

CHEMISTRY

26. (2)
27. (1)
28. (4)
29. (3)
30. (1)
31. (4)
32. (4)
33. (3)
34. (3)
35. (1)
36. (1)
37. (4)
38. (4)
39. (2)
40. (3)
41. (4)
42. (2)
43. (3)
44. (3)
45. (4)
46. (433)
47. (200)
48. (60)
49. (26)
50. (3)

MATHEMATICS

51. (2)
52. (3)
53. (4)
54. (1)
55. (1)
56. (4)
57. (1)
58. (1)
59. (4)
60. (1)
61. (2)
62. (1)
63. (3)
64. (4)
65. (2)
66. (1)
67. (4)
68. (3)
69. (2)
70. (3)
71. (30)
72. (78)
73. (25)
74. (5)
75. (5)

SECTION-I (PHYSICS)

1. **(4)**

$\vec{v} = 2t^2\hat{i} + 3t\hat{j}$

$\vec{a} = \frac{dv}{dt} = 4t\hat{i} + 3\hat{j}$

Where $t = 2$ sec $\Rightarrow \quad \vec{a} = 8\hat{i} + 3\hat{j}$

$\therefore \quad \vec{F} = \overrightarrow{ma}$

$\Rightarrow \quad x = 8$

[8 April, 2023 (Shift-I)]

2. **(3)**

$mvl = mv'l + \left(\frac{1}{3}Ml^2\right)\omega \quad \ldots(i)$

$v + v' = \omega l \quad \ldots(ii)$

$\Rightarrow \quad \frac{6v}{5l} = \omega$

$\frac{1}{2}\times\left(\frac{1}{3}Ml^2\right)\omega^2 = Mg\frac{l}{2}(1-\cos\theta)$

$1-\cos\theta = \frac{12v^2}{25g}$

$\cos\theta = -0.2$

$\theta = 180 - 78 = 102°.$

[3 Sep, 2020 (Shift-I)]

3. **(1)**

$W = \int \vec{F}.\vec{dr}$

$\left[x^3 + y^2\right]_{(2,1)}^{(3,2)}$

= 22 J.

[24 June, 2022 (Shift-I)]

4. **(2)**

At the surface, escape velocity = $\sqrt{\frac{2GM}{R_e}}$

Applying conservation of energy:

$\Rightarrow \quad \frac{-GMm}{R_e} + \frac{1}{2}m\left(\sqrt{\frac{2GM}{R_e}}\right)^2 = \frac{-GMm}{r} + \frac{1}{2}mv^2$

$\Rightarrow \quad \frac{1}{2}mv^2 = \frac{GMm}{r} \Rightarrow v^3 = 2\frac{2GM}{r}$

$\Rightarrow \quad v = \sqrt{\frac{2GM}{r}}$

$\Rightarrow \quad = \frac{dr}{dt} = \sqrt{\frac{2GM}{r}}$

$t = \frac{R_e}{3}\sqrt{\frac{2R_e}{GM}}\left(\left(1+\frac{h}{R_e}\right)^{3/2} - 1\right)$

Time t taken by body to reach at height R_e.

[22 July, 2021 (Shift-II)]

5. **(2)**

$\frac{150-T}{\left(\frac{R}{2}\right)} + \frac{80-T}{\frac{R}{2}} = \frac{T-50}{R}$

T= 102°C.

Heat current in PD = $\frac{\Delta T}{R} = \frac{52}{10} = 5.2\,\text{W}$

[27 Aug, 2021 (Shift-I)]

6. **(4)**

Resultant force on charge q_0

$F_R = \left\{-\frac{2kQq_0x}{(a^2+x^2)^{\frac{3}{2}}}\right\}$

From Newtons second law, $a = \frac{F}{m}$

$\therefore \quad a = \left\{-\frac{2kQq_0x}{(a^3)m}\right\}$

$\omega^2 = \left\{\frac{2kQq_0}{(a^3)m}\right\}$

$T = 2\pi\sqrt{\frac{a^3m}{2kQq_0}} = \sqrt{\frac{4\pi^2a^3m\times 4\pi\varepsilon_0}{2Qq_0}} = \sqrt{\frac{8\pi^3\varepsilon_0 ma^3}{Qq_0}}$

[27 July, 2022 (Shift-I)]

7. **(4)**

Total flux through the cube $(q/\varepsilon^0)\times 1/8 = q/8\varepsilon^0$

Total flux through one "outer" face of the cube

$(q/8\varepsilon_0)\times 1/3 = q/24\varepsilon_0$

[Because there is flux only through 3 faces]

$\phi_T = (q/25\varepsilon_0 + q/24\varepsilon_0)\times\frac{1}{2}$ [half of each face is shaded]

$\phi_T = q/24\varepsilon_0$

$\phi_{\text{remaining}} \quad \frac{q}{8\varepsilon_0} - \frac{q}{24\varepsilon_0} = \frac{q}{12\varepsilon_0}$

[25 Feb, 2021 (Shift-I)]

8. **(1)**

$y = \left(\frac{2\varepsilon C}{3}\right)$

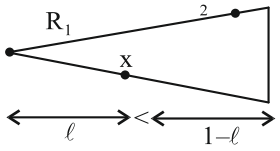
$x = \left(\frac{\varepsilon^2 C}{3}\right)$

$\frac{x}{y} = \left(\frac{\varepsilon}{2}\right)$

[12 Jan, 2019 (Shift-I)]

9. **(4)**

$\frac{R'}{R_1} = \frac{R'}{R_2} \Rightarrow R_1 = R_2 = \frac{R}{2}$



Total resistance = R

$$\int_{\text{Wire}} dR = \int_0^1 \frac{k}{l^{\frac{1}{3}}}.d\ell = k\left[\frac{3}{2}l^{\frac{2}{3}}\right]_0^1 = \frac{3k}{2}$$

$$R_1 = \frac{R}{2} = \frac{3k}{4} \Rightarrow \int_0^{Q.} \frac{k}{l^{\frac{1}{3}}}.d\ell = \frac{3k}{4}$$

$$= k\left[\frac{3}{2}l^{\frac{2}{3}}\right]_0^{l'} = \frac{3k}{4}$$

$$l'^{\frac{2}{3}} = \frac{1}{2}, l' = \left(\frac{1}{2}\right)^{\frac{3}{2}}$$

[12 Jan, 2019 (Shift-I)]

10. (2)

$$\phi = \int_x^{x+d} \frac{B_0 x^2 d}{a^2} dx$$

$$\phi = \frac{B_0 d}{3a^2}\left((x+d)^3 - x^3\right)$$

$$\mathrm{E_{ind}} = \frac{d\phi}{dt} = \mathrm{E_{ind}} = \frac{B_0 d^3 v_0}{a^2}$$

[16 March, 2021 (Shift-II)]

11. (3)

Applying Snell's law

$$2.5 \times \sin 45^\circ = n \times \sin 90^\circ$$

$$\Rightarrow \quad 2.5 \times \frac{1}{\sqrt{2}} = n = \frac{5}{2\sqrt{2}}$$

[26 July, 2022 (Shift-II)]

12. (1)

Velocity perpendicular to the river flow

$= \sqrt{8^2 - 6^2} = 2\sqrt{7}$ km/hr

Time = 2 hr.

[16 March, 2021 (Shift-II)]

13. (2)

For C-R_2 circuit the capacitive reactance is

$$X_C = \frac{1}{\omega C} = \frac{1}{100 \times \frac{2}{\sqrt{3}} \times 10^{-3}} = 5\sqrt{3}\Omega$$

For L-R_1 circuit branch, inductive reactance is

$$X_L = \omega L = 100 \times \frac{\sqrt{3}}{10} = 10\sqrt{3}\Omega$$

Peak value of I in the circuit is $\frac{20}{Z_{eff}}$

Peak value of $I = \sqrt{\frac{7}{3}}$ A.

[12 Jan, 2019 (Shift-II)]

14. (4)

Time constant, $\tau = CR = \left(\frac{k\varepsilon_0 A}{d}\right)\left(\rho\frac{d}{A}\right)[Q = d]$

$= (\rho K \varepsilon_0)$

For discharging RC-circuit

$$q = q_0 e^{-t/\tau}$$

$$I = \frac{dq}{dt} = \frac{q_0}{\tau} e^{-t/\tau}$$

$$\therefore \quad I = \frac{q_0}{\rho k \varepsilon_0} e^{-t/\tau}$$

Therefore, leakage current,

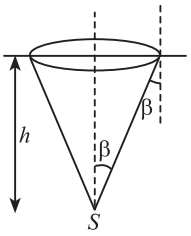
$$I_{\max} = \frac{q_0}{\rho k \varepsilon_0} = 10 \text{ mA}$$

[26 Aug, 2021 (Shift-I)]

15. (2)

Suppose the cone of light formed on the surface of water, as shown in the figure β is the crtical angle

$$\sin\beta = \frac{3}{4}, \cos\beta = \frac{\sqrt{7}}{4}$$



We know, solid angle $d\Omega = 2\pi R^2(1-\cos\beta)$

Percentage of light $= \frac{2\pi R^2(1-\cos\beta)}{4\pi R^2} \times 100$

$$= \frac{(1-\cos\beta)}{2} \times 100 = \left(\frac{4-\sqrt{7}}{8}\right) \times 100 \approx 17\%$$

Here percentage of light coming out of surface remains constant with depth.

[9 Jan, 2020 (Shift-II)]

16. (3)

$$I = I_0 \cos^2(\omega t) \Rightarrow I_{av} = \frac{I_0}{2}$$

$$\therefore \quad E = \frac{I_0}{2} \times A \times (\Delta t)$$

$$\Delta t = \frac{2\pi}{\omega} = \frac{2 \times 3.14}{15.7} = 0.4\text{s}$$

$$\therefore \quad E = \frac{\left(\frac{20}{7}\right)}{2} \times 7 \times 10^{-4} \times 0.4 = 4 \times 10^{-4} \text{ J}$$

[4 Sep, 2020 (Shift-I)]

17. (1)

$\omega = 5 \times 10^{14} \times 2\pi$, $f = 5 \times 10^{14}$

$$\lambda = \frac{C}{f} = \frac{3 \times 10^8}{5 \times 10^{14}} = 6000\text{Å}$$

Energy of photon = $\frac{12400}{6000} = 2.07\ eV$

From Einstein's equation

$KE_{max} = E - \phi$

$eV_s = E\ \phi$

$eV_s = 2.07 - 1.5$

$eV_s = 0.57$

$eV_s = 0.57\ eV$

$V_s = 0.57\ V = 0.57$ volt

[9 April, 2019 (Shift-I)]

18. (1)

In 1st orbit radius is R_o

$\left(R \propto \frac{n^2}{Z}\right)$

$R_3 = 9R_0$

$\left(R \propto \frac{n^2}{Z}\right)$

$R_{5th} = 25R_0 = \left(\frac{25R}{9}\right)$

[31 Jan, 2023 (Shift-II)]

19. (1)

$Q = (m_A - m_B - m_D) \times c^2$

$= (238.05079 - 234.04363 - 4.00260)$

$= (0.00456c^2).u$

$0.00456c^2.931.5\frac{MeV}{c^2} \qquad [\because 1u = 931\frac{MeV}{c^2}]$

$\Rightarrow$ 4.25 MeV

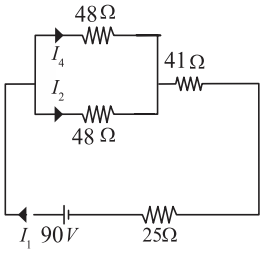
Kinetic energy of B = $\frac{4}{238} \times 4.25$

= 0.07 MeV

[13 April 2023 (Shift-I)]

20. (2)

As D_2 is reverse biased. So reduced circuit



$R_{Total} = 90\Omega$

$I_1 = \frac{90}{90} = 1A$

Using KCl $I_1 = I_2 + I_4$

$I_2 = I_4 = \frac{1}{2} = 0.5A$

[10 April 2023 (Shift-II)]

21. (4)

The equation of wave at any time t can be written as

$y = \frac{1}{1 + (x - vt)^2}$

Comparing above equation, $y = \frac{1}{1 + (x-4)^2}$

At t = 1s, We get $v = 4\ m/s$

[20 July, 2021 (Shift-I)]

22. (36)

$B + \left(2\pi \times \frac{\sqrt{3}}{2}R\right) \times \frac{\sqrt{3}}{2}T = \frac{4}{3}\pi R^3 \times d$

$B = \left(\frac{5\pi R^3}{24}\right) \times \rho g$

$R = \sqrt{\frac{36T}{(32d - 5\rho)g}}$

[9 Jan, 2020 (Shift-II)]

23. (37)

$mk^2 = \frac{1}{12}mL^2 + \frac{mL^2}{25}$

$k = \sqrt{\frac{37}{300}}L$

[7 Jan, 2020 (Shift-I)]

24. (27)

$\frac{1}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For *A* to *B*, $n_1 = 2$, $n_2 = 3$

$\frac{1}{\lambda_1} = R(1)^2\left(\frac{1}{(2)^2} - \frac{1}{(3)^2}\right) = R\left(\frac{5}{36}\right)$ .....(i)

For *B* to *C*, $n_1 = 3$, $n_2 = 4$

& $\frac{1}{\lambda_2} = R(1)^2\left[\frac{1}{(2)^2} - \frac{1}{(4)^2}\right] = R\left(\frac{3}{16}\right)$ …(ii)

Dividing (ii) by (i)

$\frac{\lambda_1}{\lambda_2} = \frac{3/16}{5/36} = \frac{27}{20}$

$\therefore \quad n = 27$

[15 April, 2023 (Shift-I)]

25. (7)

$I = 4I_0\cos^2\left(\frac{\Delta\phi}{2}\right)$ and path different

$= \frac{\lambda}{2\pi} \times$ phase diff.

For path difference $\frac{\lambda}{6}$, phase difference $\phi = \frac{\pi}{3}$

$\Rightarrow \quad I_1 = 3I_0$

For path difference λ, path difference $\phi = 2\pi$

$\Rightarrow \quad I_2 = 4I_0$

$= \frac{I_1 + I_2}{I_0} = 7$

[30 Jan, 2023 (Shift-II)]

SECTION-II (CHEMISTRY)

26. (2)

On passing alpha particles through the gold foil, some of them get deflected at small angle, few of them bounced back at 180° and most of them went straight

[27 July, 2021 (Shift-II)]

27. (1)

As PV = nRT, where n & P are constant

$$\frac{P}{nR} = k = T/V_m \qquad \ldots(i)$$

Isobaric expansion $T \propto V_m$ (graph $\rightarrow$ D)

$$T = \frac{k}{1/V_m} \rightarrow (\text{Graph B})$$

[12 Jan, 2019 (Shift-II)]

28. (4)

The following reaction takes place:

$$Cu^{2+} + 2KI \rightarrow \underset{\text{Unstable}}{CuI_2}\downarrow + 2K^+$$

I^- is strong R.A it reduces Cu^{2+} to Cu^+

$$2CuI_2 \rightarrow \underset{\text{(White) 'P'}}{Cu_2I_2}\downarrow + I_2$$

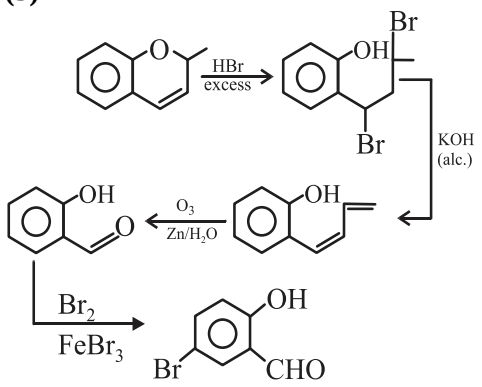
$$KI + I_2 \rightarrow K^+I_3^- (\text{Brown solution})$$

$$I_3^- \rightleftharpoons I_2 + I^-$$

$$KI_3 + Na_2S_2O_3 \rightarrow KI + \underset{(Q)}{Na_2S_4O_6}$$

[31 Jan, 2023 (Shift-I)]

29. (3)



[2 Sept, 2020 (Shift-I)]

30. (1)

The electronic configuration with respect to given atomic numbers are as follow,

49 – belongs to p-block

39 – belongs to d-block

55 – belongs to s-block

97 – belongs to f-block

[30 Jan, 2023 (Shift-I)]

31. (4)

$CH_3Cl > CH_2Cl_2 > CHCl_3 > CCl_4$

[2 Sept, 2020 (Shift-I)]

32. (4)

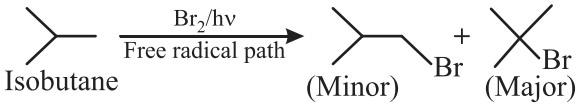
As we move down the group, acidic character of oxide decreases.

$SiO_2 > GeO_2 > SnO_2 > PbO_2$

[6 April, 2023 (Shift-II)]

33. (3)

The given reaction occurs as:



[26 Aug, 2021 (Shift-I)]

34. (3)

The expression for lowering of vapour pressure

$= p^0 - p = p^0.\chi_{solute}$

Given: $p^0 = 35$ mmHg

$$\chi_{solute} = \frac{n_{glucose}}{n_{glucose} + n_{water}}$$

$$\therefore \Delta p = 35 \times \frac{1.80/180}{\frac{1.80}{180} + \frac{540}{18}} = 35 \times \frac{.01}{.01 + 30} = 35 \times \frac{.01}{30.01}$$

$= .012$ mm Hg

[10 April, 2019 (Shift-I)]

35. (1)

$$E^o_{cell} = E^o_{Cu^{2+}/Cu} - E^o_{Zn^{2+}/Zn} = 1.1V$$

So, if Eext = 1.1V, no e^- will flow.

At $E_{ext} > 1.1V$, cell act as electrolytic cell and electron will flow from Cu to Zn.

At $E_{ext} < 1.1V$, cell act as electrochemical cell, so Zn dissolves at anode and Cu at Cathode.

[4 Sept, 2020 (Shift-I)]

36. (1)

Rate constant depends on temperature as per Arrhenius equation:

$K = Ae^{-Ea/RT}$

As T increases, rate constant increases exponentially

[1 Sept, 2021 (Shift-II)]

37. (4)

Bi is a metal.

[26 June, 2022 (Shift-II)]

38. (4)

Ce^{4+} ($4f^0$) = colourless (no f·f transition)

[8 April, 2019 (Shift-I)]

39. (2)

$CuSO_4 \cdot 5H_2O \Rightarrow [Cu(H_2O)_4]SO_4 \cdot H_2O$

In $CuSO_4 \cdot 5H_2O$, Cu is co-ordinated with 4 water molecule and fifth water molecule is hydrogen bonded.

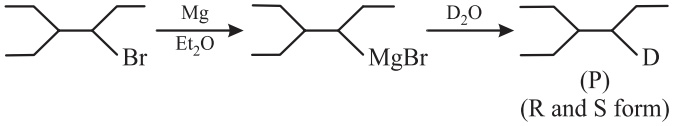
So, secondary valency = 4

No. of hydrogen bond per molecule = 1

[18 March, 2021 (Shift-II)]

40. (3)

The reaction occurs as:



(P)

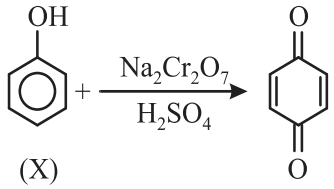
(R and S form)

[29 July, 2022 (Shift-II)]

41. (4)

X is phenol with the given empirical formula as only aromatic compounds burns with sooty flame.

The reaction takes place as:



(X)

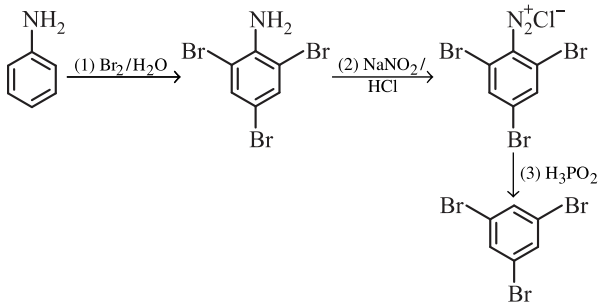
[31 Jan, 2023 (Shift-I)]

42. (2)

Presence of E.W.G. on benzene ring will increase the reactivity of CHO group towards nucleophilic addition.

[2 Sept, 2020 (Shift-I)]

43. (3)



[28 Jun, 2022 (Shift-I)]

44. (3)

Glucose and galactose both are monosaccharides.

Barfoed test:

In this test, copper acetate in dilute acid is reduced in 30 seconds by monosaccharides while disaccharides takes several minutes.

[27 July, 2021 (Shift-I)]

45. (4)

If any component eluted first then it means that its R_f value is low and hence its interaction with stationary phase is weaker

[10 April, 2023 (Shift-I)]

46. (433)

E° value for spontaneous process must be +ve

$$Fe^{3+} + Cu \rightarrow Cu^{2+} + Fe^{2+}$$

$$E^o_{cell} = E^o_{cathode} - E^o_{anode}$$

$$= (0.77 - 0.337)V = 0.433V = 433 \times 10^{-3}\ V$$

[25 June, 2022 (Shift-I)]

47. (200)

For first order reactions,

$t = 1/K \ln[A_0]/[A]$

At 44min,

$[A_0] = a_0$

$[A] = 0.6\ a_0$

For 90% decomposition,

$[A_0] = a_0$

$[A] = 0.1\ a_0$

Hence,

$$\frac{t_1}{t_2} = \frac{\frac{1}{K}\ln\frac{a_0}{0.6a_0}}{\frac{1}{K}\ln\frac{a_0}{0.1a_0}}$$

$$\frac{44}{t_2} = \frac{\ln\frac{10}{6}}{\ln 10}$$

$$\frac{44}{t_2} = \frac{\log 10 - \log 6}{\log 10}$$

$$\frac{44}{t_2} = \frac{1 - 0.78}{1}$$

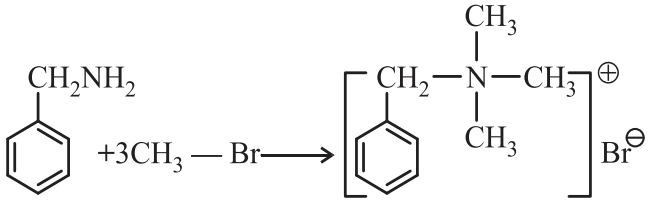
$$\Rightarrow \frac{44}{t_2} = 0.22$$

$$\Rightarrow t_2 = \frac{44}{0.22} = 200\,\text{min}$$

[30 Jan, 2023 (Shift-II)]

48. (60)

Step 1:



3 moles of bromomethane is required to form 1 mole of benzyl trimethyl ammonium bromide

Step 2:

To calculate no. of moles of benzyl trimethyl ammonium bromide present in 46 g

M.wt of $(C_{10}H_{16}NBr)$

$= 12 \times 10 + 1\times 16 + 1 \times 14 + 1 \times 80 = 230$ u

No. of moles in 46 g = $\frac{46}{230}$ = 0.2 mole

For 1 mole product $\rightarrow$ 3 mole (CH_3Br) required
So for 0.2 mole product $\rightarrow$ 0.6 mole CH_3Br is required

[18 March, 2021 (Shift-I)]

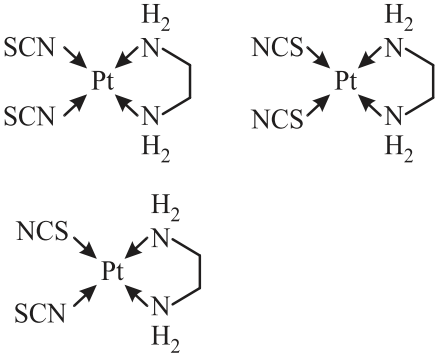
49. (26)

Mass of Na^+ in 50 mL = 140 × 50 = 7000mg
23000 mg of Na^+ = 85000 mg of $NaNO_3$ (1 mole $NaNO_3$ contains 1 mole Na^+)

$\therefore$ 7000 mg Na^+ will be present in $\frac{85000}{23000} \times 7000$
= 26g

[26 Feb, 2021 (Shift-II)]

50. (3)



Total 3 isomers are possible.

[4 Sep, 2020 (Shift-I)]

SECTION-III (MATHEMATICS)

51. (2)

$P = \{a \in \mathbb{R}: [a + 2] + [a + 4] \le 2\}$

$[a + 2] + [a + 4] \le 2$

$\Rightarrow [a] + 2 + [a] + 4 \le 2$

$\Rightarrow 2[a] \le -4 \Rightarrow [a] \le -2 \Rightarrow a < -1$

$a \in (\infty, -1)$ ….(i)

$$Q = \left\{ a \in R : 3^a \left(\sum_{r=1}^{\infty} \frac{3}{10^r} \right)^{a-3} < 3^{-3a} \right\}$$

$$\Rightarrow 3^a \cdot \left(\frac{3 \cdot \frac{1}{10}}{1 - \frac{1}{10}} \right)^{a-3} < 3^{-3a}$$

$$\Rightarrow 3^a \cdot \left(\frac{1}{9} \right)^{a-3} < 3^{-3a}$$

$$\Rightarrow \left(\frac{1}{9} \right)^{a-3} < 3^{-5a+3}$$

$\Rightarrow 3^{6-2a} < 3^{3-5a}$

$\Rightarrow 6 - 2a < 3 - 5a$

$\Rightarrow 3a < -3 \Rightarrow a < -1 \Rightarrow a \in (-\infty, -1)$ …(ii)

From (i) & (ii)
$P = Q$

[6 April, 2023 (Shift-I)]

52. (3)

Given: $28x^2 - 62x + 6m = 0$

$\Rightarrow 14x^2 - 31x + 3m = 0 \begin{matrix} \nearrow \alpha \\ \searrow \beta \end{matrix}$

$\alpha + \beta = \frac{31}{14}$ .....(i),

$\alpha\beta = \frac{3m}{14}$ ....(ii)

Also we have,
$38x^2 - 38x + 2m = 3x^2 + 15x - 2m$

$\Rightarrow 35x^2 - 53x + 4m = 0 \begin{matrix} \nearrow \alpha \\ \searrow \gamma \end{matrix}$

$\alpha + \gamma = \frac{53}{35}$ ....(iii)

$\alpha\gamma = \frac{4m}{35}$ ....(iv)

Now dividing (ii) from (iv)

$$\frac{\beta}{\gamma} = \frac{\frac{3m}{14}}{\frac{4m}{35}} = \frac{3m}{14} \times \frac{35}{4m} = \frac{3 \times 35}{14 \times 4} = \frac{15}{8}$$

$\Rightarrow \beta = \frac{15}{8}\gamma$

Now subtracting (iii) from (i)

$$\beta - \gamma = \frac{31}{14} - \frac{53}{35} = \frac{1085 - 742}{14 \times 35} = \frac{343}{490} = \frac{7}{10}$$

$\Rightarrow \frac{15}{8}\gamma - \gamma = \frac{7}{10}$

$\Rightarrow \frac{15\gamma - 8\gamma}{8} = \frac{7}{10} \Rightarrow \frac{7\gamma}{8} = \frac{7}{10}$

$\Rightarrow \gamma = \frac{8}{10} = \frac{4}{5} \Rightarrow \gamma = \frac{4}{5}$

$\therefore \beta = \frac{15}{8} \times \frac{4}{5} = \frac{3}{2}$

$\therefore \alpha = \frac{31}{14} - \beta = \frac{31}{14} - \frac{3}{2} = \frac{31 - 21}{14} = \frac{5}{7}$

$m = \frac{14}{3}\alpha\beta = \frac{14}{3} \times \frac{5}{7} \times \frac{3}{2} = 5$

Now $\alpha + m = \frac{5}{7} + 5 = \frac{5+35}{7} = \frac{40}{7}$

$\gamma + \beta = \frac{4}{5} + \frac{3}{2} = \frac{8+15}{10} = \frac{23}{10}$

Required equation:

$x^2 - \left(\frac{40}{7} + \frac{23}{10}\right)x + \frac{40}{7} \times \frac{23}{10} = 0$

$\Rightarrow \quad x^2 - \frac{561}{70}x + \frac{92}{7} = 0$

$\Rightarrow \quad 70x^2 - 561x + 920 = 0$

[29 Jan, 2023 (Shift-I)]

53. (4)

$\left(x^2 + \frac{1}{x^2}\right) - 5\left(x + \frac{1}{x}\right) + 8 = 0$

$\Rightarrow \quad \left[\left(x + \frac{1}{x}\right)^2 - 2\right] - 5\left(x + \frac{1}{x}\right) + 8 = 0$

$\Rightarrow \quad \left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right) + 6 = 0$

Let $\left(x + \frac{1}{x}\right)^2 - 5\left(x + \frac{1}{x}\right) + 6 = 0$

Let $x + \frac{1}{x} = k$

$\Rightarrow \quad k^2 - 5k + 6 = 0$

$\Rightarrow \quad k^2 - 2k - 3k + 6 = 0$

$\Rightarrow \quad k(k-2) - 3(k-2) = 0$

$\Rightarrow \quad (k-2)(k-3) = 0$

$\therefore \quad k = 2, 3$

$x + \frac{1}{x} = 2, 3$

$x^2 - 2x + 1 = 0, \; x^2 - 3x + 1 = 0$

Total 3 solutions

[24 Jan, 2023 (Shift-II)]

54. (1)

$a + 6d = 3, \qquad \ldots.(1)$

$Z = a(a + 3d)$

$= (3 - 6d)(3 - 3d)$

$= 18d^2 - 27d + 9$

Differentiating with respect to d

$\Rightarrow \quad 36d - 27 = 0$

$\Rightarrow \quad d = \frac{3}{4}$, from (1) $a = \frac{-3}{2}$, (Z = minimum)

Now, $S_n = \frac{n}{2}\left(-3 + (n-1)\frac{3}{4}\right) = 0$

$\Rightarrow \quad n = 5$

Now,

$n! - 4a_{n(n+2)} = 120 - 4(a_{35})$

$= 120 - 4(a + (35 - 1)d)$

$= 120 - 4\left(\frac{-3}{2} + 34.\left(\frac{3}{4}\right)\right)$

$= 120 - 4\left(\frac{-6+102}{4}\right)$

$= 120 - 96 = 24$

[31 Jan, 2023 (Shift -II)]

55. (1)

We have $\frac{z_1}{z} + (1-z)\bar{z}z_1 = i\bar{z} + 1$

$\Rightarrow \quad z_1\left(\frac{1}{z} + (1-z)\bar{z}\right) = i\bar{z} + 1$

$\Rightarrow \quad z_1 = \frac{i\bar{z}+1}{(1-z)\bar{z} + \frac{1}{z}} = \frac{1 + i(2-i)}{(2-i)(1-2-i) + \frac{1}{2+i}}$

$= \frac{1 + 2i - i^2}{(2-i)(-1-i) + \frac{1}{2+i} \times \frac{(2-i)}{(2-i)}}$

$= \frac{2(1+i)}{(i-2)(i+1) + \frac{2-i}{4+1}}$

$= \frac{2(1+i)}{\left[i^2 - 2i + i - 2\right] + \frac{2-i}{5}}$

$= \frac{2(1+i) \times 5}{-5(i+3) + (2-i)}$

$= \frac{10(1+i)}{-5i - 15 + 2 - i} = \frac{10(1+i)}{-13 - 6i}$

$= \frac{10(1+i)}{-(13+6i)} \times \frac{13-6i}{13-6i}$

$= \frac{10\left[13 + 13i - 6i - 6i^2\right]}{-\left[13^2 - 36i^2\right]}$

$= \frac{-10[19 + 7i]}{169 + 36}$

$= \frac{-10}{205}[19 + 7i]$

$\Rightarrow \quad \frac{205z_1}{10} = -[19 + 7i]$

$\frac{41z_1}{2} = -(19 + 7i)$

$\therefore \quad 41z_1 = -38 - 14i$

[20 Jan, 2023 (Shift-I)]

56. (4)

Let $z = x + iy$

We have

$|z - 1| = 3|z - 2|$

$\Rightarrow \quad \sqrt{(x-1)^2 + y^2} = 3\sqrt{(x-2)^2 + y^2}$

$\Rightarrow \quad (x-1)^2 + y^2 = 9(x-2)^2 + 9y^2$

$\Rightarrow \quad x^2 - 2x + 1 + y^2 = 9x^2 - 36x + 36 + 9y^2$

$\Rightarrow \quad 8x^2 + 8y^2 - 34x + 35 = 0$

$\Rightarrow \quad x^2 + y^2 - \frac{34}{8}x + \frac{35}{8} = 0$

$P(a, b) = P\left(\frac{17}{8}, 0\right)$

$\therefore \quad k = \sqrt{\left(\frac{17}{8}\right)^2 - \frac{35}{8}}$

$\Rightarrow \quad K = \frac{3}{8}$

$64\left(\frac{17}{8} + 0 + 4 \cdot \frac{9}{64}\right) = 172$

[1 Feb, 2023 (Shift-I)]

57. (1)

Vowels: *I*, *E*, *E*, *E*, *E*

Consonants: *N*, *N*, *N*, *D*, *D*, *P*, *C*

$\boxed{I \quad E \quad E \quad E \quad E}$ 3*N*, 2*D*, *P*, *C*

Number of required words = $\frac{8!}{3!\,2!} \times \frac{5!}{4!} = 16800$

$\therefore \quad m = 16800$

And words start with *P*

P _____

So, 4*E*, 3*N*, 2*D* (repetition are there)

$n = \frac{11!}{4!3!2!} = 138600$

$\therefore \quad m + n = 16800 + 138600$

$= 155400$

[8 April, 2023 (Shift-I)]

58. (1)

Passing of lines L_1 and L_2 are $(\lambda, 2, 1)$ & $\left(\sqrt{3}, 1, 2\right)$

$$S.D. = \frac{\begin{vmatrix} \sqrt{3}-\lambda & -1 & 1 \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix}}{\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 1 & 1 \\ 1 & -2 & 1 \end{vmatrix}}$$

$1 = \left|\frac{\sqrt{3}-\lambda}{\sqrt{3}}\right|$

$\lambda = 0, \lambda = 2\sqrt{3}$

$m = \frac{0 + 2\sqrt{3}}{\sqrt{3}}$

$= 2$

[6 April, 2023 (Shift-II)]

59. (4)

$$\sum_{r=1}^{26}\left(\frac{2r}{(2r)!\left[51-(2r-1)!\right]}\right)$$

$$= \sum_{r=1}^{26}\left(\frac{2r}{2r(2r-1)!\left[51-(2r-1)!\right]}\right)$$

$$= \sum_{r=1}^{26}\left(\frac{1}{(2r-1)!\left[51-(2r-1)!\right]}\right)$$

$$= \sum_{r=1}^{26}\left({}^{51}C_{2r-1} \cdot \frac{1}{51!}\right)$$

$$= \frac{1}{51!}\left({}^{51}C_1 + {}^{51}C_3 + ... + {}^{51}C_{51}\right)$$

$$= \frac{1}{51!} \times 2^{50}$$

$$= \frac{2^{50}}{51!}$$

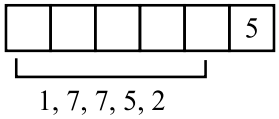
[1 Feb, 2023 (Shift-I)]

60. (1)

1, 7, 7, 5, 5, 2

Total number of six digits number = $\frac{6!}{2!\,2!}$

$\Rightarrow \quad n(s) = 180$



1, 7, 7, 5, 2

Total six digits number that are

Divisible by $5 = \frac{5!}{2!} = 60$

Required probability = $\frac{60}{180} = \frac{1}{3}$

[26 Feb, 2021 (Shift-I)]

61. (2)

$8x^2 - 8ax + 10a = 0$

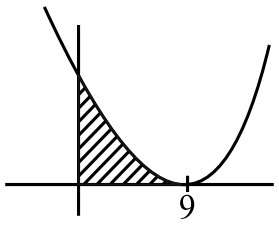
For rational roots, *D* must be a perfect square

$D = 64a^2 - 320a$

$= 64\,(a^2 - 5a)$

at $a = 9$, *D* is perfect square.

$\therefore$ Parabola: $y = x^2 - 18x + 81 = (x - 9)^2$



Area = $\int_0^9 (x-9)^2\,dx$

= 243 sq. units

[30 Jan, 2023 (Shift-II)]

62. (1)

$$X=\begin{bmatrix}1 & 3 & 3^2\\ \frac{1}{3} & 1 & 3\\ \frac{1}{3^2} & \frac{1}{3} & 1\end{bmatrix}$$

$$X^2=\begin{bmatrix}1 & 3 & 3^2\\ \frac{1}{3} & 1 & 3\\ \frac{1}{3^2} & \frac{1}{3} & 1\end{bmatrix}\begin{bmatrix}1 & 3 & 3^2\\ \frac{1}{3} & 1 & 3\\ \frac{1}{3^2} & \frac{1}{3} & 1\end{bmatrix}$$

$$X^2=\begin{bmatrix}3 & 3^2 & 3^3\\ 3\cdot\frac{1}{3} & 3 & 3^2\\ 3\cdot\frac{1}{3^2} & 3\cdot\frac{1}{3} & 3\end{bmatrix}=3\begin{bmatrix}1 & 3 & 3^2\\ \frac{1}{3} & 1 & 3\\ \frac{1}{3^2} & \frac{1}{3} & 1\end{bmatrix}$$

$\Rightarrow \quad X^2 = 3X$

Similarly

$X^3 = 3^2X$, ……

$X + X^2 + X^3 + ….. + X^{20}$

$= X + 3X + 3^2X + …… + 3^{19}X$

$$= X\left(1+3+3^2+.....+3^{19}\right)=X\left(\frac{3^{20}-1}{3-1}\right)$$

$$=\frac{\left(3^{20}-1\right)X}{2}$$

[29 June, 2022 (Shift-I)]

63. (3)

$$Q=\frac{1}{3!}\begin{bmatrix}6 & -6\\ 12 & 24\end{bmatrix}=\frac{1}{6}\begin{bmatrix}6 & -6\\ 12 & 24\end{bmatrix}=\begin{bmatrix}1 & -1\\ 2 & 4\end{bmatrix}$$

Now

$P = Q^T$

$$\Rightarrow \quad P=\begin{bmatrix}1 & 2\\ -1 & 4\end{bmatrix}$$

$|P| = 4 + 2 = 6$

Now $P^{-1}=\frac{adJP}{|P|}=\frac{1}{6}\begin{bmatrix}4 & -2\\ 1 & 1\end{bmatrix}$

$$\Rightarrow \quad P^{-1}=\begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}$$

We have $P^{-1} = aI + bP$

$$\begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}=\begin{bmatrix}a & o\\ o & a\end{bmatrix}+\begin{bmatrix}b & 2b\\ -b & 4b\end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix}\frac{2}{3} & -\frac{1}{3}\\ \frac{1}{6} & \frac{1}{6}\end{bmatrix}=\begin{bmatrix}a+b & 2b\\ -b & a+4b\end{bmatrix}$$

Comparing both side

$$a+b=\frac{2}{3},\ b=-\frac{1}{6}$$

[27 July, 2022 (Shift-II)]

64. (4)

sin12° + cos78° – cos18°

= sin12° + sin(90° – 78°) – sin(90° – 18°)

= sin12° + sin12° – sin72°

= sin12° – 2cos42°· sin30°

$$=\sin 12^\circ-2\cdot\sin\left(90^\circ-42^\circ\right)\cdot\frac{1}{2}$$

= sin12° – sin48°

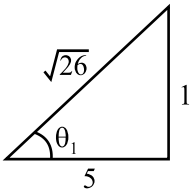
= – 2 cos30° · sin18°

$$=-2\times\frac{\sqrt{3}}{2}\times\frac{\sqrt{5}-1}{4}=\frac{\sqrt{3}\left(1-\sqrt{5}\right)}{4}$$

[25 June, 2022 (Shift-I)]

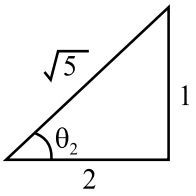
65. (2)

$$\alpha=2\cos^{-1}\left(\frac{5}{\sqrt{26}}\right)+\sin^{-1}\left(\frac{1}{\sqrt{5}}\right)+2\tan^{-1}\frac{1}{8}$$



$$\cos\theta_1=\frac{5}{\sqrt{26}},\tan\theta_1=\frac{1}{5}\Rightarrow\theta_1=\tan^{-1}\frac{1}{5}$$

and



$$\sin\theta_2=\frac{1}{\sqrt{5}},\tan\theta_2=\frac{1}{2}\Rightarrow\theta_2=\tan^{-1}\frac{1}{2}$$

Then

$$\alpha=2\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{2}+2\tan^{-1}\frac{1}{8}$$

$$=2\left(\tan^{-1}\frac{1}{5}+\tan^{-1}\frac{1}{8}\right)+\tan^{-1}\frac{1}{2}$$

$$=2\left[\tan^{-1}\left(\frac{\frac{1}{5}+\frac{1}{8}}{1-\frac{1}{40}}\right)\right]+\tan^{-1}\frac{1}{2}$$

$$=2\tan^{-1}\frac{1}{3}+\tan^{-1}\frac{1}{2}$$

$$= \tan^{-1}\left(\frac{\frac{2}{3}}{1-\frac{1}{9}}\right) + \tan^{-1}\frac{1}{2}$$

$$= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{2}$$

$$= \tan^{-1}\left(\frac{\frac{3}{4}+\frac{1}{2}}{1-\frac{3}{8}}\right)$$

$$= \tan^{-1}\left(\frac{\frac{5}{4}}{\frac{5}{8}}\right) = \tan^{-1}(2) = \cot^{-1}\left(\frac{1}{2}\right)$$

Now $\tan(\tan^{-1}(2)) + \cot(\tan^{-1}2)$

$$= 2 + \cot\left(\cot^{-1}\left(\frac{1}{2}\right)\right)$$

$$= 2 + \frac{1}{2} = \frac{5}{2}$$

[26 July, 2022 (Shift-I)]

66. (1)

Let $S = \sum_{k=0}^{119}\left(\left[\sqrt{k+1}\right]\right)$

$$= \left[\sqrt{1}\right] + \left[\sqrt{2}\right] + \left[\sqrt{3}\right] + ... + \left[\sqrt{100}\right] + ... + \left[\sqrt{120}\right]$$

$$= \underbrace{1+1+1}_{3\text{ times}} + \underbrace{2+2+....+2}_{5\text{times}} + \underbrace{3+3+...+3}_{7\text{ times}} + ... + \underbrace{10+10+.....+10}_{21\text{times}}$$

$= 3 \times 1 + 5 \times 2 + 7 \times 3 + \ldots. + 10 \times 21$

$$= \sum_{r=1}^{10}(2r+1)r$$

[13 April, 2023 (Shift-II)]

67. (4)

$$\sqrt{2+2\cos\theta-\sin^2\theta} = \sqrt{1+2\cos\theta+1-\sin^2\theta}$$

$$= \sqrt{1+2\cos\theta+\cos^2\theta}$$

$= 1 + |\cos\theta|$

So,

$$f(\theta) = \begin{cases} \left(1+|\cos\theta|\right)^{\frac{a}{\cos\theta}} & , \quad 0 < \theta < \frac{\pi}{2} \\ b & , \quad \theta = \frac{\pi}{2} \\ e^{\frac{\tan 4\theta}{\tan 6\theta}} & , \quad \theta > \frac{\pi}{2} \end{cases}$$

Given $\lim\limits_{\theta\to\frac{\pi^+}{2}} f(\theta) = \lim\limits_{\theta\to\frac{\pi^-}{2}} f(\theta) = f\left(\frac{\pi}{2}\right)$

i.e. $f(\theta)$ is continuous at $x = \frac{\pi}{2}$

Now $\lim\limits_{\theta\to\frac{\pi^+}{2}} f(\theta) = \lim\limits_{\theta\to\frac{\pi^+}{2}} e^{\frac{\tan 4\theta}{\tan 6\theta}}$

$$= \lim_{\theta\to\frac{\pi^+}{2}} e^{\left(\frac{\sin 4\theta}{\cos 4\theta}\times\frac{\cos 6\theta}{\sin 6\theta}\right)} = e^{2/3}$$

$$\lim_{\theta\to\frac{\pi^-}{2}} \left(1+|\cos\theta|\right)^{\frac{a}{\cos\theta}} = e^a$$

and $f\left(\frac{\pi}{2}\right) = b$

so, $e^{2/3} = e^a = b$

$$a = \frac{2}{3},\ b = e^{2/3}$$

$$\Rightarrow \quad 3a = 2,\ \log_e b = \log_e e^{2/3} = \frac{2}{3}$$

$3(a + \ln b) = 3a + 3\ln b$

$= 2 + 2 = 4$

$x^2 - (2 + 4)x + 8 = 0$

$\Rightarrow \quad x^2 - 6x + 8 = 0$

[25 Jan, 2023 (Shift-II)]

68. (3)

$$\lambda = \frac{f(\lambda) + f'''(3)}{f''(2) - \lambda f'(1) + \lambda^2}$$

$\Rightarrow \quad f(\lambda) = \lambda^3 - \lambda^2 f'(1) + \lambda f''(2) - f'''(3),\ \lambda \in \mathbb{R}$

Let $f'(1) = A,\ f''(2) = B,\ f'''(3) = C$

then

$f(\lambda) = \lambda^3 - A\lambda^2 + B\lambda - C$

$f'(\lambda) = 3\lambda^2 - 2A\lambda + B \Rightarrow f'(1) = 3 - 2A + B$

$f''(\lambda) = 6\lambda - 2A \Rightarrow f''(2) = 12 - 2A$

$f'''(\lambda) = 6 \Rightarrow f'''(3) = 6$

$\Rightarrow \quad \lambda^3 - A\lambda^2 + B\lambda - C = \lambda^3 - \lambda^2(3 - 2A + B) + \lambda(12 - 2A) - 6$

comparing both side

$A = 3 - 2A + B \quad \Rightarrow \quad 3A - B = 3 \quad ....(i)$

$B = 12 - 2A \quad \Rightarrow \quad 2A + B = 12 \quad .....(ii)$

and $C = 6$ $\qquad 5A = 15 \ \therefore A = 3$

$\therefore\ B = 6$

$f(\lambda) = \lambda^3 - 3\lambda^2 + 6\lambda - 6$

$f(x) = x^3 - 3x^2 + 6x - 6$

$f(0) = -6,\ f(1) = -2,\ f(3) = 12,\ f(2) = 2$

$\log_{10}|f(0) + f^2(1) + f^3(2) + f(3)|$

$= \log_3|-6 + 4 + 8 + 12| = \log_3|18|$

$= \log_3|2 \times 9|$

$= \log_3 2 + \log_3 3^2$

$= \log_3 2 + 2\cdot\log_3 3$

$= \log_3 2 + 2$

[24 Jan, 2023 (Shift-II)]

69. (2)

Let

$$I=\int\left(1-\cos^2 x\right)\cos x\left(\sin^4 x+\sin^2 x+1\right)\cdot\left(2\sin^4 x+3\sin^2 x+6\right)^{1/2}dx$$

$$=\int\sin^2 x\cos x\left(\sin^4 x+\sin^2 x+1\right)\cdot\left(2\sin^4 x+3\sin^2 x+6\right)^{1/2}dx$$

Now Let $\sin x = t$

$\cos x\,dx = dt$

The given integral becomes

$$I=\int\left(t^6+t^4+t^2\right)\sqrt{2t^4+3t^2+6}\;dt$$

$$=\int\left(t^5+t^3+t\right)\sqrt{2t^6+3t^4+6t^2}\,dt$$

substitute $2t^6 + 3t^4 + 6t^2 = y$

$\Rightarrow 12\,(t^5 + t^3 + t)\,dt = dy$

Hence,

$$\frac{1}{12}\int\sqrt{y}dy=\frac{1}{18}y^{\frac{3}{2}}+c=\frac{1}{18}\left(2t^6+3t^4+6t^2\right)^{\frac{3}{2}}+c$$

$\Rightarrow \quad 18I = (2t^6 + 3t^4 + 6t^2)^{3/2} + 18c$

$= (2\sin^6 x + 3\sin^4 x + 6\sin^2 x)^{3/2} + c'$

[25 Feb, 2021 (Shift-I)]

70. (3)

$$\left(2\sqrt{x+1}\right)\frac{dy}{dx}+\frac{\left(2\sqrt{x+1}\right)}{x}\cdot y=1$$

$$\Rightarrow\quad \frac{dy}{dx}+\frac{1}{x}\cdot y=\frac{1}{2\sqrt{x+1}}$$

$$I.F=e^{\int\frac{1}{x}dx}=e^{\ln x}=x$$

Solution: $xy=\int\frac{x}{2\sqrt{x+1}}dx$

Now Let $x + 1 = t^2$

$\Rightarrow \quad dx = 2t\,dt$

$$\Rightarrow\quad xy=\int\frac{t^2-1}{2t}\cdot 2tdt$$

$$\Rightarrow\quad dy=\frac{t^3}{3}-t+c$$

$$\Rightarrow\quad xy=\frac{(x+1)^{3/2}}{3}-\sqrt{x+1}+c$$

$x = 3, y = 2$

$$\Rightarrow\quad 6=\frac{(4)^{3/2}}{3}-\sqrt{4}+c$$

$$\Rightarrow\quad c=6-\frac{8}{3}+2=8-\frac{8}{3}=\frac{16}{3}$$

$$\Rightarrow\quad yx=\frac{(x+1)^{3/2}}{3}-\sqrt{x+1}+\frac{16}{3}$$

[31 Jan, 2023 (Shift-I)]

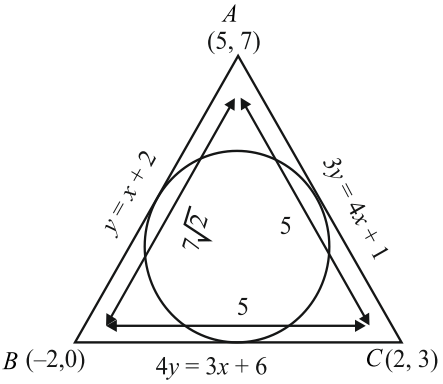
71. (30)

$7x + 14 = 7y$, $8y - 6x = 12$

$\Rightarrow \quad y = x + 2 \qquad \Rightarrow 4y = 3x + 6$

and $9y - 12x = 3$

$\Rightarrow \quad 3y = 4x + 1$



$$(h,k)=\left(\frac{5\times5+5(-2)+14\sqrt{2}}{10+7\sqrt{2}},\frac{35+21\sqrt{2}}{10+7\sqrt{2}}\right)$$

$$\therefore\quad h+k=\frac{50+35\sqrt{2}}{10+7\sqrt{2}}=\frac{5\left(10+7\sqrt{2}\right)}{10+7\sqrt{2}}=5$$

Now

$$AB^2+BC^2+CA^2=\left(7\sqrt{2}\right)^2+5^2+5^2=148$$

$$\therefore\quad AB^2+BC^2+CA^2+2=150=30\times5=30(h+k)$$

[30 Jan, 2023 (Shift-I)]

72. (78)

Ellipse: $9x^2 + 25y^2 = 225$

$$\Rightarrow\quad \frac{x^2}{25}+\frac{y^2}{9}=1$$

$$e=\sqrt{1-\frac{9}{25}}=\sqrt{\frac{25-9}{25}}=\sqrt{\frac{16}{25}}=\frac{4}{5}$$

foci $\equiv\left(\pm5\cdot\frac{4}{5},0\right)\equiv(\pm4,0)$,

Latus Rectum $\frac{2\times9}{5}=\frac{18}{5}$

Hyperbola:

$$\frac{x^2}{4}-\frac{y^2}{\left(\frac{a^2}{9}\right)}=1$$

$$e=\sqrt{1+\frac{\left(\frac{a^2}{9}\right)}{4}}$$

$$e=\sqrt{1+\frac{a^2}{36}}$$

Focus $=\left(2\cdot\sqrt{1+\frac{a^2}{36}},0\right)=(4,0)$

$$2\sqrt{1+\frac{a^2}{36}}=4$$

$$\sqrt{1+\frac{a^2}{36}}=2$$

$$1+\frac{a^2}{36}=4$$

$a^2 = 108$

Sum of latus rectum now hyperbola

$L = \frac{18}{5} + 12$ $\quad\quad$ $\frac{x^2}{4} - \frac{y^2}{\left(\frac{108}{9}\right)} = 1$

$L = \frac{18+60}{5}$ $\quad\quad$ $\frac{x^2}{4} - \frac{y^2}{12} = 1$

$5L = 78$ $\quad\quad$ L.R $\Rightarrow 2 \cdot \frac{12}{2} = 12$

[27 June, 2022 (Shift-I)]

73. (25)

Area $= \frac{1}{2}\left|\overrightarrow{BD} \times \overrightarrow{AC}\right| = 18$

$$\overrightarrow{AC} \times \overrightarrow{BD} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -3 & -3 \\ 8 & 5 & -\lambda \end{vmatrix}$$

$= (3\lambda + 15)\hat{i} - \hat{j}(0 + 24) + \hat{k}(0 + 24)$

$\overrightarrow{AC} \times \overrightarrow{BD} = (3\lambda + 15)\hat{i} - 24\hat{j} + 24\hat{k}$

$= \sqrt{(3\lambda + 15)^2 + (24)^2 + (24)^2} = 36$

$= \lambda^2 + 10\lambda + 9 = 0$

$\Rightarrow \lambda = -1, -9$

$|\lambda| \le 5 \Rightarrow \lambda = -1$

$15 - 10\lambda = 15 + 10 = 25$

[1 Feb, 2023 (Shift-I)]

74. (5)

$X = \frac{24}{5}; \sigma^2 = \frac{194}{25}$

Let first four observation be x_1, x_2, x_3, x_4

Here, $\frac{x_1 + x_2 + x_3 + x_4 + x_5}{5} = \frac{24}{5}$...

Also, $\frac{x_1 + x_2 + x_3 + x_4}{4} = \frac{7}{2}$

$\Rightarrow \quad x_1 + x_2 + x_3 + x_4 = 14$

Now from eqn-1

$x_5 = 10$

Now, $\sigma^2 = \frac{194}{25}$

$\frac{x_1^2 + x_2^2 + x_3^2 + x_4^2 + x_5^2}{5} - \frac{576}{25} = \frac{194}{25}$

$\Rightarrow \quad x_1^2 + x_2^2 + x_3^2 + x_4^2 = 54$

Now, variance of first 4 observations

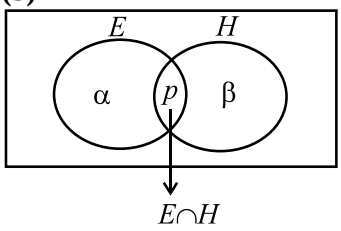
$$Var = \frac{\sum_{i=1}^{4} x_i^2}{4} - \left(\frac{\sum_{i=1}^{4} x_i}{4}\right)^2$$

$= \frac{54}{4} - \frac{49}{4} = \frac{5}{4}$

$m = \frac{5}{4} \Rightarrow 4m = 5$

[1 Feb, 2023 (Shift-II)]

75. (5)



$\alpha + p = 75$ $\quad\quad$ .....(i)

$\beta + p = 40$ $\quad\quad$ ....(ii)

$\alpha + \beta + p = 100$ $\quad\quad$ ....(iii)

From (i), (ii) and (iii)

$P = 15$, $\alpha = 60$ and $\beta = 25$

[6 April, 2023 (Shift-II)]

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